Enhancing Precision Drug Recommendations via In-Depth Exploration of Motif Relationships

Chuang Zhao, Hongke Zhao, Xiaofang Zhou, *Fellow, IEEE*, and Xiaomeng Li, *Member, IEEE*

***Abstract*—Making accurate and safe clinical decisions for patients has long been a challenging task. With the proliferation of electronic health records and the rapid advancement of technology, drug recommender systems have emerged as invaluable aids for healthcare professionals, offering precise and secure prescriptions. Among prevailing methods, the exploration of motifs, defined as substructures with specific biological functions, has largely been overlooked. Nevertheless, the substantial impact of the motifs on drug efficacy and patient diseases implies that a more extensive incorporation could potentially improve the recommender systems. In light of this, we introduce *DEPOT*, an innovative drug recommendation framework developed from a motif-aware perspective. In our approach, we employ chemical decomposition to partition drug molecules into semantic motif-trees and design a structure-aware graph transformer to capture motif collaboration. This innovative practice preserves the topology knowledge and facilitates perception of drug functionality. To delve into the dynamic correlation between motifs and disease progression, we conduct a meticulous investigation from two perspectives: repetition and exploration. This comprehensive analysis allows us to gain valuable insights into the drug turnover, with the former focusing on reusability and the latter on discovering new requirements. We further formulate a historical weighting strategy for drug-drug interaction (DDI) objective, enabling adaptive control over the trade-off between accuracy and safety criteria throughout the training process. Extensive experiments conducted on four data sets validate the effectiveness and robustness of *DEPOT*.**

***Index Terms*—Drug discovery, molecular representation learning, recommender system.**

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I. Introduction

With the advancement of deep learning technology and the availability of vast medical data, artificial intelligence (AI) has emerged as a powerful ally in medical decision-making [1]. Among the diverse range of applications, drug recommendation emerges as an increasingly crucial field. Its main goal revolves around comprehending the inherent interplay among diseases, diagnostic procedures, and medications, ultimately facilitating the personalized prescription of treatments tailored to individual patient conditions–an aspect of healthcare that holds exceptional significance [2]. Unlike conventional recommendation algorithms [3], [4], [5], drug recommendation encounters distinct challenges associated with safety requirements and personalized suggestions [6], [7]. Notable examples in this field include Baidu's Medical Brain, Berkeley Lights, and Google Health, all utilizing AI-driven drug recommender systems, as exemplified in [8], [9].

Existing drug recommendations can be categorized into three main genres: rule-based, instance-based, and longitudinal-based. The early rule-based approaches initially depend on experienced physicians manually formulating rules to guide the diagnostic process [10]. Nonetheless, this method has proven to be time-consuming, labor-intensive, and reliant on extensive expert knowledge and a substantial rule database for effective decision-making. The instance-based method draws inspiration from collaborative filtering algorithms, seeking to uncover co-occurrence relationships between drugs and symptoms [11]. However, its efficacy in discerning individual patient needs may be limited, primarily relying on commonalities observed in the broader population. More recently, longitudinal approaches have emerged, incorporating patient history to a greater extent [12]. This longitudinal observation and analysis offer a comprehensive grasp of the patient's condition, facilitating the provision of precise diagnosis, as depicted in Fig. 1(a).

While making substantial progress, these methods suffer from two common limitations. The first limitation arises from the insufficient exploration of drug substructures, referred to as motifs, despite these motifs embodying the essence of drug functionality, as exemplified in Fig. 1(b). For example, in the case of nonsteroidal anti-inflammatory drugs (NSAIDs), the *phenyl ring* and *acetic acid group* (two common motifs) synergistically form a specific pharmacophore. This pharmacophore engages with the target enzyme cyclooxygenase, leading to the inhibition of prostaglandin synthesis, thereby conferring anti-inflammatory, analgesic, and antipyretic effects [13].

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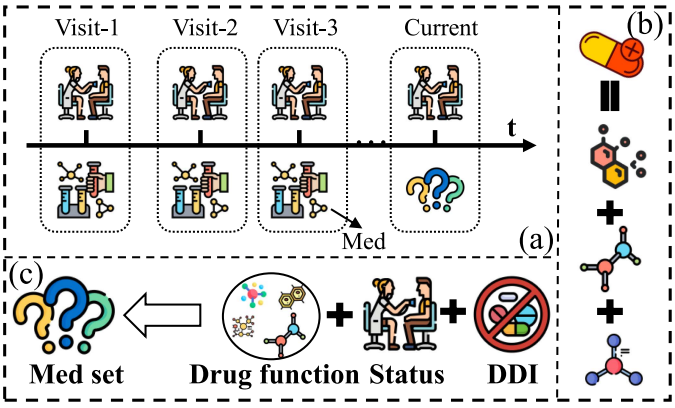


Fig. 1. (a) Denotes that patient visits and prescription processes are longitudinally tracked, with each medical (Med) prescription entailing a set of drugs. (b) Illustrates drug is composed of multiple motifs, and their collaboration constitutes the drug's function. (c) Implies drug recommendations need to take into account patient status, drug function, and drug-drug interaction(DDI) simultaneously.

However, traditional methods typically represent drugs as SMILES (Simplified Molecular Input Line Entry System) sequences [14] or atomic connectivity graphs [15], often overlooking the comprehensive exploration and modeling of these intricate collaborations among motifs. Although a few studies have touched upon motifs [12], they fall short of capturing the intricate structural interactions between them, thereby impeding the accuracy of drug representation and recommendation. The second limitation is that current approaches lack explicit modeling of the dynamic relationship between drug motifs and the evolution of a patient's medical condition, even though capturing this relationship is crucial, as evidenced in cancer treatment [16], [17]. For instance, specific motifs in anti-cancer drugs interact with signaling pathways within tumor cells, effectively inhibiting their growth, and necessitating the continuous administration of certain drugs over a specified period. However, disease progression can lead to mutations and drug resistance [18], [19], compromising the interaction between drug motifs and tumor cells and calling for the exploration of new drug motifs as replacements. Therefore, it is crucial to shift focus towards effectively integrating motif collaboration and precisely capturing the dynamic interplay between drug motifs and disease progression, as depicted in Fig. 1(c).

To address the aforementioned challenges, we present a novel framework to *enhance precision* **D***rug* **r**E*commendation via in-depth ex*P*loration of m*O*tif rela*T*ionships*, namely ***DEPOT***. First, we harness specific chemical knowledge [20] to transform each drug molecule into a **motif-tree**, and associate it with the proposed **structure-aware graph transformer** that effectively captures long-distance collaboration effects between motifs. This motif-level perspective alleviates the low semantic richness of previous atomic-level aggregation as well as facilitates a more comprehensive understanding of structural dependencies and topology knowledge within the molecule. Second, to model the dynamic relationship between patient disease conditions and drug motifs, we adopt two key perspectives. On the one hand, we design a **repeat encoder** that evaluates the probability of re-prescribing a drug with certain motifs, taking into account the patient's disease progression and the functional characteristics of the drug in their historical records. This enables us to assess the degree to which disease progression necessitates the continuation of specific drug functionalities. On the other hand, we introduce an **explore encoder** that computes probability using all available motifs, facilitating the discovery of novel drug options. Additionally, we establish a **discriminate encoder** to adjust the balance between the two components. Third, we delve deep into the Drug-Drug Interaction (DDI) effect [21] and leverage **historical DDI records** to reconstruct the DDI loss function in a nonlinear fashion. This innovative approach not only stabilizes the training process but also strengthens safety considerations in drug recommendation by accounting for the historical DDI information.

In summary, our work has the following contributions:

- To the best of our knowledge, *DEPOT* stands as the pioneering drug recommendation framework dedicated to mining motifs relations. Unlike previous approaches, we not only consider structural information and collaboration among drug motifs but also delve into the dynamic relationship between motifs and disease progression from two distinct perspectives: repetition and exploration.
- We offer a fresh perspective on tackling the drug-drug interaction problem. We introduce historical DDI effect sequences for the first time, presenting an innovative solution to address safety concerns in drug recommendation with stability and efficiency.
- We conduct rigorous experimental validation on four extensive public data sets, showcasing substantial advancements achieved by our algorithm in terms of accuracy, safety, and practicality. We have released the code and data sets[1] for follow-up and reproducibility.

The remainder of this paper is structured as follows, Section II reviews the closet related work done. Section III formalizes the definition of drug recommendation and elaborates each sub-module of *DEPOT*. In Section IV, we present the experimental results and various robust tests and hyper-parameter tests. Finally, we draw a conclusion in Section V.

II. Related Work

To provide readers with a comprehensive understanding of the advancements in the field and establish the connections and distinctions between our work and other studies, this section presents a systematic review of the most pertinent literature in the domains of drug recommendation and molecular characterization.

A. Drug Recommendation

Drug recommendation, as a challenging sub-problem in the field of smart healthcare, serves as an essential tool to assist doctors in making prescriptions and alleviating their workload. In contrast to traditional recommender systems [22], [23], drug recommender systems face a dual challenge of *recommending*

[1]https://github.com/xmed-lab/DrugRec

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target sets of varying sizes and *adhering to stringent safety criteria* [6], [24], [25].

In the early stages of drug recommendation research, manual rule-based methods were commonly employed. For instance, techniques like K-freq [26] and K-Near [27] relied on selecting drugs based on their symptom frequency or the drug set used in previous visits. However, these approaches heavily depended on the expertise of the doctors and lacked flexibility in adapting to individual patient needs. Inspired by recommender systems used in e-commerce [3], instance-based methods were introduced to enhance the efficiency and accuracy of drug recommendation. These practices exploited the co-occurrence relationship between drugs and patients, akin to collaborative filtering algorithms. For example, MedRec [11] constructed a knowledge graph linking symptoms and drugs and utilized link prediction techniques. ECC [28] formulated the recommendation task as a multi-label classification problem and employed chained ensemble classifiers to improve precision. While effective, these static methods fail to account for the patient's historical status, leading to a significant information deficit. Subsequently, researchers explored drug recommendation from a longitudinal perspective, taking into account patient history and temporal dependencies among drugs. GAMENet [27] employed recurrent neural networks (RNNs) to model the patient's historical states and combined electronic health records (EHRs) with drug-drug interaction graphs to explore co-occurrence relationships among drugs. SafeDrug [6] utilized the same longitudinal characterization module and further integrated drug atomic graphs and fingerprints into the framework to enhance recommendation performance. COGNet [29] retrieved the patient's historical diagnoses and drug prescriptions, and mined their relationship with the current diagnosis to generate personalized recommendations. Safety considerations also gained attention to promote the preference for safer drug combinations. For instance, 4SDrug [30] and MoleRec [12] incorporated small and safe principals into the training process of the recommendation model, utilizing a mixed penalty loss to ensure that the generated drug combinations carried lower risk.

Despite decent performance, these methods often overlook crucial motifs information. One relevant work in this regard is MoleRec [12], which primarily focuses on the inclusion relationship between drugs and substructures. While this approach captures some aspects of motifs information, it fails to fully explore the structural intricacies between motifs and their dynamic connections with changes in patient diseases. Our proposed framework, DEPOT, takes this into account by introducing motif trees, allowing us to capture essential motifs information for more robust molecular characterization. In addition, we conduct dynamic analysis from the perspectives of repetition and exploration, as well as introducing historical DDI into safety consideration, two dimensions not available in other works.

B. Molecular Characterization

Effective characterization of drugs is crucial for enhancing the performance of drug recommendations, as it needs to capture complex drug-disease and drug-drug relationships. Currently, there are two main streams of molecular modeling methods: sequence-based models and graph-based representation models [31], [32], [33], [34].

Sequence-based models drew inspiration from successful approaches in natural language processing. These models could capture the statistical features and semantic information of molecular SMILES sequences, enabling the understanding of molecular composition and properties. For instance, recurrent neural networks or transformers could be employed to model SMILES sequences and make predictions on molecular properties [35], [36], [37]. On the other hand, graph-based models represented molecules as graphs, where atoms or groups are nodes, and chemical bonds are edges [38], [39]. This approach incorporated non-euclidean geometry data mining and took into account the connections, topological relationships, and chemical bond information between atoms. Graph convolutional neural networks (GCNs) or graph attention networks (GATs) were commonly utilized to model molecular graphs and predict molecular properties [40], [41]. In recent years, efforts have been made to unify sequence-based and graph-based modeling methods by representing molecular graphs as atomic sequences and utilizing positional encoding to incorporate coordinate information [42], [43]. This paradigm not only benefits from the ability of sequence models to capture long-distance dependencies but also excels in capturing complex collaborations between nodes when processing graph data.

In our study, we draw inspiration from molecular characterization and innovatively employ various chemical decomposition methods to divide drug molecules into meaningful motifs. Distinguished from previous work, we additionally consider the structural associations between these motifs and integrate them into the graph transformer model. This higher-level and finer-grained approach aids in comprehending the motif collaboration and subtle connections between drugs and diseases, thereby improving the performance of drug recommendations.

III. Proposed Method

We first outline the operational process of *DEPOT* and then provide a detailed explanation of each sub-module's mechanism. Finally, we discuss the training and optimization strategies for the model.

A. Problem Formulation

EHR Data: Suppose we have a electronic health record data set with $\mathcal{N}$ patients, where each patient visit record is denoted as $\mathcal{X}^{(k)} = (\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}, \ldots, \mathbf{x}_{\mathcal{N}_k}^{(k)})$. Here, $k \in \mathcal{N}$ represents the patient index, and $\mathcal{N}_k$ represents the total number of visits for that patient. Each visit $\mathbf{x}_i^{(k)}$ can be represented as a triplet: $\mathbf{x}_i^{(k)} = (\mathbf{d}_i^{(k)}, \mathbf{p}_i^{(k)}, \mathbf{m}_i^{(k)})$, where $\mathbf{d}, \mathbf{p}$, and $\mathbf{m}$ denote multi-hot diagnosis, procedure, and medication records, respectively. The variables $\mathbf{d}_i^{(k)} \in \{0,1\}^{|\mathcal{D}|}$, $\mathbf{p}_i^{(k)} \in \{0,1\}^{|\mathcal{P}|}$, and $\mathbf{m}_i^{(k)} \in \{0,1\}^{|\mathcal{M}|}$, where $\mathcal{D}, \mathcal{P}, \mathcal{M}$ represent the set of each object, and $|\cdot|$ represents the size of the set. For each drug, it can be decomposed into a set of motifs through molecular

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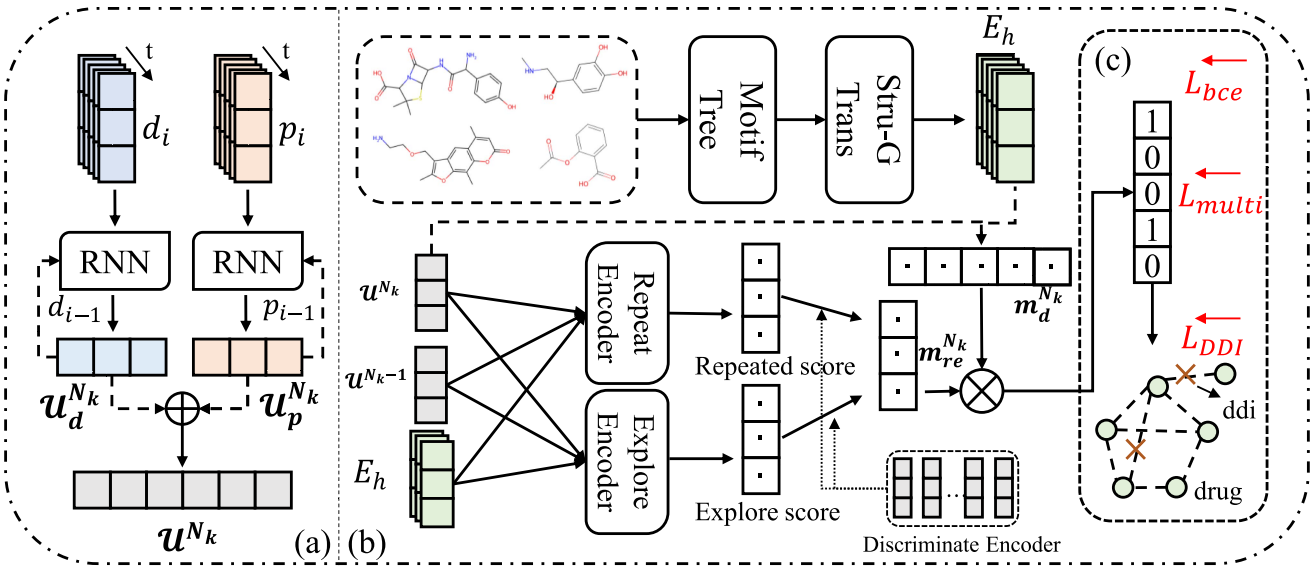


Fig. 2. Overall framework of *DEPOT*. Stru-G Trans denotes structure-aware graph Transformer and X refers to removing the DDI pair. (a) Longitudinal patient representation module, which extracts patients' historical diagnoses and procedures. (b) Motif-aware drug representation module, employed to extract relationships between motifs and drug functions, as well as motifs and disease progression. (c) Drug recommendation module, considering both accuracy and safety objectives.

TABLE I
MATHEMATICAL NOTATIONS

Notations	Descriptions
$\mathcal{X}$	electronic health record
$\mathcal{D}$, $\mathcal{P}$, $\mathcal{M}$,$\mathcal{I}$	diag, proc, drug, and motif set
$\mathbf{d}$, $\mathbf{p}$, $\mathbf{m}$	multi-hot vector of diag, proc, drug, and motif
$\mathbf{s}$, $\mathbf{s}^{deg}$, $\mathbf{s}^{rec}$	one-hot vector of motif, degree, and recency
$\mathbf{x}$	concatenation of medical codes
$\mathcal{G}$	DDI graph
$\mathcal{V}$	vertex set same as $\mathcal{M}$
$\mathcal{E}$, $\mathbf{A}$	edge set and adjacency matrix of $\mathcal{G}$
$\mathbf{e}_*$	medical embedding of medical codes $*$
$\mathbf{u}^*$	patient embedding at $*$ visit
$\mathbb{E}_*$	embedding tables of medical codes $*$
$\mathbf{z}$	patient disease progression
$\mathbf{p}_*$	probability vector of mode $*$
$\mathbf{S}$	drug-motif adjacent matrix
$\hat{\mathbf{o}}$	recommendation probability vector for drugs
β, α	hyperparameters for accuracy and safety loss
η	target ddi value
κ	mean value of historical ddi rate

decomposition [44], i.e., $d_i = \{s_{i,1}, \ldots, s_{i,j}\}$, where d_i denotes drug i, $s \in \mathcal{I}$ and $\mathcal{I}$ denotes the set of motifs for all drugs.

Safety Criterion: The drug-drug interaction (DDI) graph, denoted as $G = \{\mathcal{V}, \mathcal{E}\}$, represents the strong side effects between different drugs. Here, $\mathcal{V} = \mathcal{M} = \{m_1, \ldots, m_{|\mathcal{M}|}\}$ denotes the set of all drugs, and $\mathcal{E}$ represents the known side effects between pairs of drugs. To elucidate the connections within $\mathcal{E}$, we utilize an adjacency matrix $\mathbf{A} \in \{0,1\}^{|\mathcal{M}|\times|\mathcal{M}|}$, where $\mathbf{A}_{i,j} = 1$ if there exists a documented side effect between drug i and drug j. Recommendations that involve a higher number of drug side effects are deemed less reliable and more dangerous. Therefore, the objective is to minimize the DDI rate, which indicates a lower occurrence of side effects, in order to enhance the safety of recommendations. Now, we can provide a formal task definition of *DEPOT*.

Drug Recommendation:: Given the patient's historical electronic medical records $\mathbf{d}_{his}^{(k)} = (\mathbf{d}_1^{(k)}, \ldots, \mathbf{d}_{\mathcal{N}_k-1}^{(k)})$, $\mathbf{p}_{his}^{(k)} = (\mathbf{p}_1^{(k)}, \ldots, \mathbf{p}_{\mathcal{N}_k-1}^{(k)})$, and $\mathbf{m}_{his}^{(k)} = (\mathbf{m}_1^{(k)}, \ldots, \mathbf{m}_{\mathcal{N}_k-1}^{(k)})$, along with the current state $(\mathbf{d}_{\mathcal{N}_k}^{(k)}, \mathbf{p}_{\mathcal{N}_k}^{(k)})$ and the DDI graph G, our objective is to recommend a drug combination $\hat{\mathbf{m}}_{\mathcal{N}_k}^{(k)}$ while maintaining a low drug-drug interaction rate.

Important mathematical notes can be found in Table I.

B. *Overview of* DEPOT *Framework*

In this section, we outline the proposed drug recommendation framework *DEPOT*, whose architecture is shown in Fig. 2. The *DEPOT* model comprises three essential modules: *Longitudinal Patient Representation Module* (LPM), *Motif-aware Drug Representation Module* (MDRM), and *Drug Recommendation Module* (DRM). 1) The primary objective of LPM is to extract a meaningful representation of the patient from their historical electronic medical records. 2) MDRM is responsible for characterizing drugs using two sub-modules: *the Drug-centric Representation Module* and *the Patient-centric Representation Module*. The Drug-centric Representation Module reconstructs drug molecules into motif-trees to capture structural information and motif collaboration. The Patient-centric Representation Module explores drug functions from repeat and explore perspectives to address temporal aspects effectively. 3) DRM combines the patient representation and drug characterization to generate personalized and safe multi-label drug predictions for each patient. By incorporating the collaborative efforts of these three modules, *DEPOT* can generate the personalized and safe drug recommendations, ultimately reducing the burden on doctors and enhancing the quality and efficiency of medical decision-making.

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C. Longitudinal Patient Representation Module

Longitudinal temporal data contains richer semantic information compared to using only current visit records, facilitating a better understanding of personalized disease progression.

Embedding Layer: Patient representation is encoded using diagnosis and procedure information, which are both multi-hot variables. To project a multi-hot variable $\mathbf{d}_i$ or $\mathbf{p}_i$ for a particular visit i into a high-dimensional embedding space, i.e. $\mathbf{e}_d^i$ and $\mathbf{e}_p^i$, we sum the embeddings of the corresponding active elements. For simplicity, we have omitted the patient index (k), the same as below. Formally,

$$\mathbf{e}_d^i = \mathbf{d}_i \mathbb{E}_d, \quad \mathbf{e}_p^i = \mathbf{p}_i \mathbb{E}_p, \tag{1}$$

where embedding spaces for diagnosis and procedure are denoted as $\mathbb{E}_d$ and $\mathbb{E}_p$ respectively.

Long-term Aggregation: A patient's health status is a dynamic process influenced by various factors such as disease progression and historical treatment records. Relying solely on current symptoms may not provide a comprehensive understanding of the patient's overall condition and disease evolution. Formally, given a patient's clinical history represented as $[\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{\mathcal{N}_k}]$ and $[\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_{\mathcal{N}_k}]$, we extract the embeddings $\mathbf{e}_d$ and $\mathbf{e}_p$ and utilize the RNN architecture to capture the visit-level diagnosis and procedure information, respectively. This process can be defined as follows,

$$\begin{aligned} \mathbf{u}_d^{\mathcal{N}_k} &= \mathrm{GRU}_d\left(\mathbf{e}_d^1, \mathbf{e}_d^2, \ldots, \mathbf{e}_d^{\mathcal{N}_k}\right), \\ \mathbf{u}_p^{\mathcal{N}_k} &= \mathrm{GRU}_p\left(\mathbf{e}_p^1, \mathbf{e}_p^2, \ldots, \mathbf{e}_p^{\mathcal{N}_k}\right), \end{aligned} \tag{2}$$

where $\mathbf{u}$ denotes the hidden vector extracted by Gated Recurrent Unit (GRU) and GRU is a two-layer structure. The two vectors will be concatenated to form the patient's final longitudinal representation to query candidate drug combinations. Formally,

$$\mathbf{u}^{\mathcal{N}_k} = g\left(\mathbf{u}_d^{\mathcal{N}_k} || \mathbf{u}_p^{\mathcal{N}_k}\right), \tag{3}$$

where $g(\cdot)$ denotes the transformation matrix and $||$ is the concatenation operation.

D. Motif-Aware Drug Representation Module

In this section, we introduce two modules for enhanced drug characterization: *Drug-centric Representation Module* focuses on uncovering motif interactions within the drug, while *Patient-centric Representation Module* considers the dynamic nature of drug effects, accounting for variations based on the patient's condition over time.

1) Drug-Centric Representation Module: The motif present in a drug holds significant importance in determining its functionality [45], we therefore model drug representation in the motif-level.

Motif Tree Construction: To preserve both semantic and structural information to the fullest extent, 1) we initially perform motifs fragmentation using well-defined rules and a strategy involving chemical bond reactions, specifically employing the BRICS method [44] as an example. This process involves splitting drug molecules based on 16 predefined rules and a set of reactive bonds, while adding dummy nodes to indicate the positions of the motifs. 2) Additionally, we follow MGSSL's [45] post-processing approach to mitigate combinatorial explosion and enhance motif co-occurrence. 3) Next, we reassemble the motifs based on the positions of the dummy nodes and record the degree of each node and the frequency of each motif. Using *CNC[C@H](O)C1=CC(O)=C(O)C=C1* as an example, we first construct a molecular object based on its SMILES expression. Subsequently, by applying steps 1) 2) and 3), we derive a motif tree with a unique node set {'CN', 'CC', 'CO', 'C', 'C1=CC=CC=C1'}, where the adjacency matrix is stored in dummy nodes. Ultimately, leveraging the existing motif tree, we can extract two pieces of structural information for each motif, namely recency and degree, denoted as {'CN':2, 'CC':2, 'CO':3, 'C':1, 'C1=CC=CC=C1':1} and {'CN':2, 'CC':2, 'CO':1, 'C':3, 'C1=CC=CC=C1':3}, respectively. Degree and recency respectively refer to the number of edges connected to the motif and the number of times the motif appears in the motif tree. Integrating these two types of topology knowledge may aid the model in comprehending the significance and activity of the node in the graph structure [46], thereby perceiving the key to drug function. An intuitive example can be seen in Fig. 3.

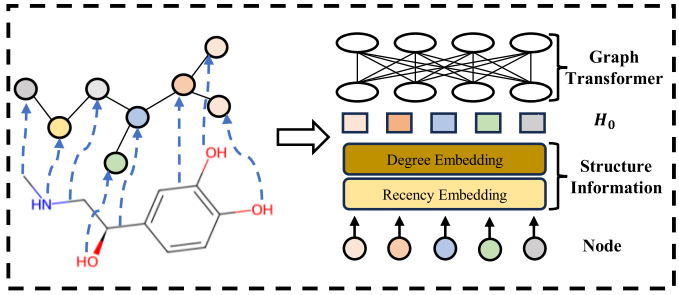


Fig. 3. Drug-centric representation module, including motif tree construction and structure-aware graph transformer. Different colored circles refer to different motifs and H_0 denotes the initial embedding.

Structure-aware Graph Transformer: Harnessing the modeling capabilities of graph transformer [20], [42] to capture intricate relationships between nodes, we employ this approach to capture molecular-level representations. Initially, we count the substructures contained in all drug molecules, recode them into a dictionary $\mathcal{I}$, and assign embedding vectors by $\mathbb{E}_s$. Meanwhile, considering the potential loss of structural information when converting topological data to sequence data, we have innovatively incorporated rich structural embeddings, including degree $\mathbb{E}_{deg}$ and recency $\mathbb{E}_{rec}$, to enhance the completeness of the representation. Formally, for a motif $\mathbf{s}_{i,j}$ contained in drug i,

$$\mathbf{e}_s^{i,j} = \mathbf{s}_{i,j}\mathbb{E}_s + \mathbf{W}_s\left(\mathbf{s}_{i,j}^{deg}\mathbb{E}_{deg} + \mathbf{s}_{i,j}^{rec}\mathbb{E}_{rec}\right), \tag{4}$$

where $\mathbf{e}_s^{i,j}$ denotes the embedding vector of $s_{i,j}$ and $\mathbf{W}_s$ is the transformation matrix for structure information. $\mathbf{s}$, $\mathbf{s}^{deg}$, and $\mathbf{s}^{rec}$ represent the one-hot encoding of s corresponding to id, degree and recency. Subsequently, we utilize a graph encoder to capture high-order complex collaborative relationships between drug molecules by incorporating the obtained semantically

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rich representations. This graph encoder consists of stacked multi-head attention mechanisms, layernorm, and feed-forward layers, ensuring a comprehensive and formal encoding process for the molecular data. Formally, for certain motifs contained in the drug molecular, the update method is as follows,

$$\mathbf{H}_i^{(0)} = \left[\mathbf{h}_{i,1}^{(0)}, \mathbf{h}_{i,2}^{(0)}, \ldots, \mathbf{h}_{i,\mathcal{N}_s^i}^{(0)}\right]^\top, \tag{5}$$

$$\mathbf{H}_i^{(l)} = \text{G-Transformer}\left(\mathbf{H}_i^{(l-1)}\right), \forall l \in \{1, 2, \ldots, D\}, \tag{6}$$

where $\mathbf{h}_{i,j}^{(0)}$ is initialized with $\mathbf{e}_s^{i,j}$, which refers to the high-level representation of the motif j contained in drug i, $\mathcal{N}_s^i$ represents the number of motifs of drug i, D represents the number of encoder layers and $\top$ is matrix transpose. Finally, without loss of generality, we use a mean pooling readout function to obtain molecular-level representations, i.e., $\mathbf{h}^i = \text{READOUT}(\mathbf{H}_i^{(D)})$, where $\mathbf{h}^i \in \mathbb{E}_h$ denotes the represenation of drug i. Next, we can compute the estimation for each drug using the patient's longitudinal status. Formally,

$$\mathbf{m}_d^{\mathcal{N}_k} = \text{LN}\left(\sigma\left(\mathbf{u}^{\mathcal{N}_k}\mathbb{E}_h^{(\top)}\right)\right), \tag{7}$$

where σ is the sigmoid activation function and LN denotes the layernorm normilization.

2) Patient-Centric Representation Module: Apart from capturing static drug functional representations, we also incorporate patient disease progression to dynamically derive patient-centric functional requirements from two angles: repetition and exploration.

Repeat Encoder: This encoder is utilized to evaluate the probability of re-prescribing a drug with certain motifs, taking into account the patient's disease progression and the functional characteristics of the drug in their historical records. We therefore first sight ingenious ways to characterize the progress of the patient's disease. Formally,

$$\mathbf{q}^u = \mathbf{u}^{\mathcal{N}_k} - \mathbf{u}^{\mathcal{N}_k - 1}, \tag{8}$$

where $\mathbf{q}^u$ represents the varying disease. Then we utilize the variational autoencoder [47] to stabilize training and alleviate noise by modeling distributions rather than directly exploiting representations. More specifically, this sampling process enables the model to incorporate uncertainty into the representation process, aligning with the nature of the medical field, where changes in disease states often manifest uncertain and dynamic characteristics [27], [48]. Formally,

$$\begin{aligned}&\mathbf{q}^u = \text{ReLU}\left(\mathbf{W}_u\mathbf{q}^u\right), \quad \boldsymbol{\mu} = \mathbf{W}_\mu\mathbf{q}^u, \quad \log\boldsymbol{\sigma} = \mathbf{W}_\sigma\mathbf{q}^u,\\ &\mathbf{z}^u = \boldsymbol{\mu} + \epsilon \odot \boldsymbol{\sigma}, \epsilon \sim \mathcal{N}(0, I),\end{aligned} \tag{9}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ represent the mean and variance, respectively, ϵ denotes the sampled random value, $\mathbf{W}$ denotes the learnable matrix, and $\mathcal{N}(0, I)$ refers to Gaussian distribution. Subsequently, we aggregate all the motifs contained in the previously prescribed drugs for a patient as a functional set $\mathcal{I}_S$ and compute the attention scores between $\mathbf{z}^u$ and each motif. Formally,

$$\mathbf{e}_r^i = \mathbf{v}_r^\top \tanh\left(\mathbf{W}_r\mathbf{e}_s^i + \mathbf{U}_r\mathbf{z}^u\right),$$

$$P(i \mid r, u, \mathcal{I}_S) = \begin{cases} \frac{\sum_i \exp(\mathbf{e}_r^i)}{\sum_{j=1}^{|\mathcal{I}_S|}\exp(\mathbf{e}_r^j)} & \text{if } i \in \mathcal{I}_S \\ 0 & \text{if } i \in \mathcal{I} - \mathcal{I}_S \end{cases}, \tag{10}$$

where $\mathbf{v}, \mathbf{U}$ are learnable vectors and $\mathbf{e}_s \in \mathbb{E}_s$. $P(a \mid b)$ signifies the probability of event a occurring given the occurrence of event b, under the assumption that the probability of selecting i is influenced by the repetition mode r, the exploited substructure $\mathcal{I}_S$, and the user's representation $\mathbf{z}^u$.

Explore Encoder: The exploration module is employed to assess the necessity of novel motifs functionalities, denoted as $i \notin \mathcal{I}_S$, thereby recommending new drugs containing such motifs. In this case, due to the progression of the patient's medical condition or changes in symptoms, there is a potential need for additional drugs with new functional properties. Hence, we perform a similar calculation of the patient's medical condition distribution but compute the probabilities associated with all substructures. Formally,

$$\mathbf{e}_e^i = \mathbf{v}_e^\top \tanh\left(\mathbf{W}_e\mathbf{e}_s^i + \mathbf{U}_e\mathbf{z}^u\right),$$

$$\mathbf{c}_e^{\mathcal{I}_S} = \sum_{i=1}^{|\mathcal{I}_S|} \frac{\exp(\mathbf{e}_e^i)}{\sum_{j=1}^{|\mathcal{I}_S|}\exp(\mathbf{e}_e^j)}\mathbf{e}_e^i, \tag{11}$$

where $\mathbf{c}$ refers to the patient state combined with motifs requirements. Considering that $\exp(-\infty) = 0$, we assume that if there exists an item in $\mathcal{I}_S$, the probability of recommending it under the explore mode is zero. Formally,

$$\mathbf{f}_i = \begin{cases} -\infty & \text{if } i \in \mathcal{I}_S \\ \mathbf{W}_e^c\mathbf{c}_e^{\mathcal{I}_S} & \text{if } i \in \mathcal{I} - \mathcal{I}_S \end{cases},$$

$$P(i \mid e, u, \mathcal{I}_S) = \frac{\exp(\mathbf{f}_i)}{\sum_{j=1}^{I}\exp(\mathbf{f}_j)}, \tag{12}$$

where $\mathbf{f}$ is equivalent to masking the used motif and e denotes the exploration mode.

Discriminate Encoder: This module calculates the personalized probability of repetition and exploration for the current patient, and the input is the patient's entire historical visit records. We exploit the attention mechanism to synthesize personalized probability distributions. Formally,

$$\mathbf{e}_{re}^i = \mathbf{v}_{re}^\top \tanh\left(\mathbf{W}_r\mathbf{e}_s^i + \mathbf{U}_r\mathbf{u}^{\mathcal{N}_k}\right),$$

$$\mathbf{c}_d^{\mathcal{I}_S} = \sum_{i=1}^{|\mathcal{I}_S|} \frac{\exp(\mathbf{e}_{re}^i)}{\sum_{j=1}^{|\mathcal{I}_S|}\exp(\mathbf{e}_{re}^j)}\mathbf{e}_{re}^i, \tag{13}$$

$$[P(r \mid u, \mathcal{I}_S), P(e \mid u, \mathcal{I}_S)] = \text{softmax}\left(\mathbf{W}_{re}^c\mathbf{c}_d^{\mathcal{I}_S}\right), \tag{14}$$

where softmax denotes the activation function, $P(r \mid u, \mathcal{I}_S)$ and $P(e \mid u, \mathcal{I}_S)$ correspond to the probabilities associated with patient needs being repetition and exploration, respectively. *Based on these three modules, we can obtain the repeat and explore probability of each patient-centric motif using the Bayesian formula.* Formally,

$$\begin{aligned}\mathbf{p}_r^u &= P\left(r \mid u, \mathcal{I}_S\right) P\left(i \mid r, u, \mathcal{I}_S\right) \mid i \in |\mathcal{I}|,\\ \mathbf{p}_e^u &= P\left(e \mid u, \mathcal{I}_S\right) P\left(i \mid e, u, \mathcal{I}_S\right) \mid i \in |\mathcal{I}|,\end{aligned} \tag{15}$$

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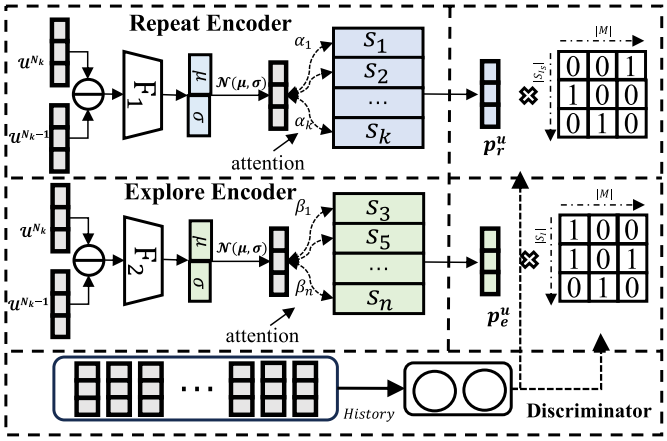


Fig. 4. Patient-centric representation module, including repeat encoder, explore encoder and discriminate encoder. The probability output vector $\mathbf{p}$ will be multiplied element-wise with the drug-motif adjacency matrix $\mathbf{S}$ to obtain the drug recommendation probability.

where $\mathbf{p}_*$ denotes the probability vector for all motifs under mode $*$. Furthermore, by incorporating the drug-motif inclusion matrix, we can obtain patient-centric probabilities for all drugs. Formally,

$$\mathbf{m}_{re}^{\mathcal{N}_k} = \mathbf{p}_r^u \mathbf{S}_{\mathcal{I}_S} + \mathbf{p}_e^u \mathbf{S}_I, \tag{16}$$

where $\mathbf{S}_{\mathcal{I}_S}$ and $\mathbf{S}_{\mathcal{I}}$ are the drug-motif adjacent matrix of $\mathcal{I}_S$ and $\mathcal{I}$, respectively. Please note that when the patient has only one historical record, we exclusively leverage the exploration aspect to discern the relationship between the motif and the user's disease progression, i.e, $\mathbf{m}_{re}^{\mathcal{N}_k} = \mathbf{p}_e^u \mathbf{S}_I$. Fig. 4 illustrates the aforementioned process in a clear and accessible manner.

E. Drug Recommendation Module

We multiply the Drug-centric retrieval probability and the Patient-centric retrieval probability to obtain the final drug recommendation probability for a patient. Formally,

$$\hat{\mathbf{o}}_{\mathcal{N}_k} = \sigma\left(\mathbf{m}_{re}^{\mathcal{N}_k} \odot \mathbf{m}_{d}^{\mathcal{N}_k}\right), \tag{17}$$

where $\odot$ denotes the Hadarmard product [23]. In general, the three sub-modules we designed not only integrate the long-term representation of patients, but also obtain recommendation results from three perspectives: functionality, reusability, and exploratory.

F. Model Training

Accuracy-oriented Loss: To minimize the gap between *DEPOT* recommendations and the groundtruth, we employ a hybrid loss comprising binary cross-entropy and multi-label margin loss for optimization. Formally,

$$\mathcal{L}_{\text{ACC}} = \gamma \mathcal{L}_{bce} + (1-\gamma)\mathcal{L}_{\text{multi}},$$

$$\mathcal{L}_{\text{multi}} = \sum_{i,j:m_{\mathcal{N}_k}^i=1, m_{\mathcal{N}_k}^j=0} \frac{\max(0, 1-(\hat{o}_{\mathcal{N}_k}^i - \hat{o}_{\mathcal{N}_k}^j))}{|M|},$$

$$\mathcal{L}_{\text{bce}} = -\sum_{i=1}^{|\mathcal{M}|}\left[m_{\mathcal{N}_k}^i \log\left(\hat{o}_{\mathcal{N}_k}^i\right) + \left(1-m_{\mathcal{N}_k}^i\right)\log\left(1-\hat{o}_{\mathcal{N}_k}^i\right)\right] \tag{18}$$

where $m_{\mathcal{N}_k}^i$ is the i-th entry of multi-hot target medication set $\mathbf{m}$ (similar define $\hat{o}_{\mathcal{N}_k}^i$ for $\mathbf{o}$), and γ is a hyper-parameter, experimentally set to 0.95.

Safety-oriented Loss: In addition to addressing accuracy concerns, we also take into account the safety aspect of drug recommendation. Follow [6], [29], using the DDI adjacency matrix $\mathbf{A}$, we design the following loss function to ensure the least side effects,

$$\mathcal{L}_{\text{DDI}} = \sum_{i=1}^{|\mathcal{M}|}\sum_{j=1}^{|\mathcal{M}|}\left(\hat{o}_{\mathcal{N}_k}^i \cdot \hat{o}_{\mathcal{N}_k}^j\right)\cdot \mathbf{A}_{ij}, \tag{19}$$

where $\hat{o}_{\mathcal{N}_k}^i \cdot \hat{o}_{\mathcal{N}_k}^j$ computes the combination probability between two drugs. This approach helps mitigate the potential increase in DDI effects within each recommendation set.

Total Loss: Given the presence of erroneous prescriptions by doctors, real Electronic Health Record data contains a certain level of Drug-Drug Interaction effect [27], [30]. Regardless of whether the predictions are correct or incorrect, they have the potential to increase the occurrence of DDIs. To address this issue, we devise a novel weighting strategy (WS) that aims to find a solution that improves both prediction accuracy and reduces the DDI effect. Formally,

$$\mathcal{L} = \alpha \mathcal{L}_{\text{ACC}} + (1-\alpha)\mathcal{L}_{\text{DDI}}, \tag{20}$$

where α is hyper-parameters to achieve a balance. In contrast to prior research that solely examines the DDI rate of the current training batch [27], we incorporate the historical DDI to promote training stability and smoothness, thereby mitigating the influence of outliers. Formally,

$$\alpha = \begin{cases} 1 & \kappa \le \eta \\ \min\left\{1, \tanh\left(\tau \frac{\eta}{\kappa - \eta}\right)\right\} & \kappa > \eta \end{cases}, \tag{21}$$

where κ is the mean of fixed-length T historical DDI rates. In this way, our optimization goal is modified to comprehensively extract historical DDI information to ensure that the average value of multiple DDIs is low, thereby ensuring the safety of the model. The overall optimization process of the algorithm is shown in Algorithm 1.

IV. Experiment

In this section, we present *data set statistics, necessary parameter settings*, and *advanced baselines*.

A. Data Sets Analysis

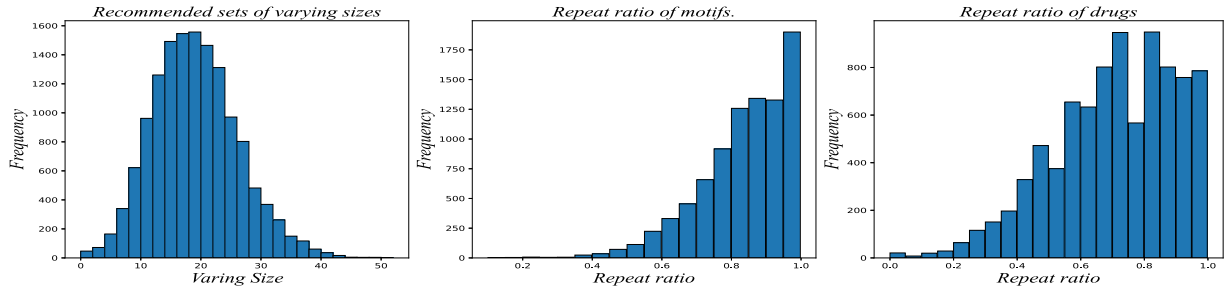
We conduct experiments using four widely used real-world Electronic Health Record data sets: *MIMIC-III* [49], *MIMIC-IV* [50], *eICU* [51], and *OMOP* [52]. The MIMIC-III data set encompasses data from adult patients in the intensive care unit between 2001 and 2012, while the MIMIC-IV data set comprises comprehensive electronic medical records and clinical information of hospitalized patients between 2008 and 2019.

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TABLE II
DATA STATISTICS

Items	MIMIC-III	MIMIC-IV	eICU	OMOP
# of patients / # of visits	6350 / 15 032	61 267 / 163 875	8600 / 18 691	743 / 3192
diag. / prod. / drug. set size	1958 / 1430 / 130	2000 / 11 055 / 130	1358 / 437 / 1411	None / 1194 / 234
avg. / max # of visits	2.3672 / 29	2.6748 / 70	2.1734 / 8	4.2961 / 27
avg. / max # of diag per visit	10.5089 / 128	8.2345 / 270	14.1709 / 106	None / None
avg. / max # of prod per visit	3.8437 / 50	2.3579 / 95	45.9671 / 144	14.0824 / 38
avg. / max # of drug per visit	11.4296 / 65	6.4889 / 72	15.9897 / 76	1.5479 / 8
total # of DDI pairs	674	800	None	None
total # of motifs	225	225	1411	234

means the number of and avg denotes the average number. Please note that the eICU and OMOP data sets do not furnish DDI pairs, and the OMOP lacks diagnose information.



(a) Distribution-set size. (b) Distribution-reuse motifs. (c) Distribution-reuse drugs.

Fig. 5. Data analysis. (a) Pertains to the distribution of drug recommendation quantities. (b) Relates to the distribution of motif reuse. (c) Denotes the distribution of drug reuse. Repeat ratio refers to the probability that the current motifs/drugs appeared in previous visits.

Algorithm 1: The Algorithm of *DEPOT*.

Input: EHR $\mathcal{X}$, DDI Graph $\mathcal{G}$
Output: Parameters Θ;
1: Random initialize model parameters Θ,
2: Data preprocessing $\mathbf{e}_d \in \mathbb{E}_d$, $\mathbf{e}_p \in \mathbb{E}_p$, $\mathbf{e}_s \in \mathbb{E}_s$
3: **while** not converged **do**
4: Sample a patient $\mathcal{X}^{(k)}$ from $\mathcal{X}$
5: **for** $t = 1$ to $\mathcal{N}_k$ **do**
6: Obtain patient representation $\mathbf{u}$ in (3)
7: Obtain drug-centric representation $\mathbf{h}$ in (6)
8: Calculate drug-centric estimation $\mathbf{m}_d$ in (7)
9: Obtain disease progression $\mathbf{z}$ in (9)
10: Obtain repeating probability of the $\mathcal{I}_S$ in (10)
11: Obtain exploring probability of the $\mathcal{I} - \mathcal{I}_S$ in (12)
12: Obtain the discrimination probability in (13)
13: Calculate patient-centric probability $\mathbf{m}_{re}$ in (16)
14: Generate the recommendation result $\hat{\mathbf{o}}$ in (17)
15: **end for**
16: Evaluate and obtain DDI rate for the patient
17: Update the parameters using the loss in (20)
18: **end while**
19: **return**Parameters Θ

The eICU aggregates data from numerous intensive care units spanning over 20 hospitals across the United States, while the OMOP data collection is grounded in an open community data standard. It is important to note that these data sets are fully anonymized and subject to privacy protection measures before our access. Table II provides detailed statistics regarding the data sets, offering insights into patient numbers, diagnosis codes, medication usage, and molecular composition.

To test our hypotheses and intuitions, we perform preliminary data analysis on MIMIC-III. As depicted in Fig. 5(a), we notice that drug recommendation scenarios require the recommendation of a collection of drugs of various sizes, rather than a single item, which represents a significant departure from traditional recommendation mechanisms. Fig. 5(b) displays the proportion distribution of historical drug reuse, while Fig. 5(c) presents the distribution of historical motif reuse. Please note that reuse here refers to the drug/motif set of the current visit that contains the drugs/motifs of previous visits. Both distributions demonstrate a right-skewed pattern, providing compelling evidence for the importance of leveraging repetition and exploration perspectives.

In a nutshell, these data analysis results serve as an empirical foundation for our research, ensuring the reliability and feasibility of our experiments.

B. Baselines

In line with the genre in related work, we have chosen several advanced baselines for comparison.

- *K-Near [27]:* Exactly the same medications recommended for the current visit as prescribed for the last visit.
- *LR [53]:* LR is a traditional recommendation technique with L1 regularization, we train a separate classifier for each class for multi-label classification.
- *ECC [28]:* ECC employs boosting-based ensemble learning for multi-label classification.
- *RETAIN [54]:* RETAIN integrates attention mechanism and RNN with the GRU architecture to represent historical electronic medical records.
- *LEAP [55]:* LEAP formulates drug recommendation as a sequential decision, utilizing a recurrent decoder and an attention mechanism to capture label dependencies.

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TABLE III
DETAILED BASELINES COMPARISON

Method	Category	Data	Drug Structure	DDI Loss
K-near	Rule	EHR	None	None
Logistic Regression	Instance	EHR	None	None
ECC	Instance	EHR	None	None
RETAIN	Longitudinal	EHR	None	None
LEAP	Longitudinal	EHR	None	None
GAMENet	Longitudinal	EHR, EHR graph, DDI graph	None	Linear
MICRON	Longitudinal	EHR, DDI graph	None	Linear
SafeDrug	Longitudinal	EHR, Molecular, DDI graph	Atom graph, Fingerprint	Linear
COGNet	Longitudinal	EHR, EHR graph, DDI graph	Atom graph	Linear
Molerec	Longitudinal	EHR, Molecular, Motif, DDI graph	Atom graph, Motif Inclusion	Non-Linear
Ours	Longitudinal	EHR, Motif, DDI graph	Motif tree	Historical + Non-Linear

The EHR graph is a correlation graph formed by diagnoses, procedures, and drugs that have appeared previously, while the DDI graph additionally records drug side effects. Molecular refers to utilizing the atomic graph, while motif refers to using substructures.

TABLE IV
PERFORMANCE COMPARISON: MIMIC-III DATA SET

Method	Jaccard ↑	F1 ↑	PRAUC ↑	DDI ↓	Avg.# of Drugs
K-near	0.4691 ± 0.0021	0.6268 ± 0.0012	0.6576 ± 0.0011	0.0762 ± 0.0007	20.56 ± 0.2836
Logistic Regression	0.4900 ± 0.0027	0.6470 ± 0.0026	0.7227 ± 0.0031	0.0774 ± 0.0006	16.47 ± 0.1091
ECC	0.4868 ± 0.0016	0.6428 ± 0.0030	0.7602 ± 0.0024	0.0806 ± 0.0011	16.01 ± 0.1107
RETAIN	0.4875 ± 0.0022	0.6469 ± 0.0027	0.7595 ± 0.0023	0.0827 ± 0.0007	19.19 ± 0.2303
LEAP	0.4672 ± 0.0023	0.6401 ± 0.0022	0.7506 ± 0.0022	0.0725 ± 0.0012	19.02 ± 0.0508
GAMENet	0.5092 ± 0.0007	0.6654 ± 0.0007	0.7675 ± 0.0007	0.0806 ± 0.0007	25.98 ± 0.2509
MICRON	0.5170 ± 0.0012	0.6722 ± 0.0018	0.7628 ± 0.0016	0.0719 ± 0.0011	18.87 ± 0.1323
SafeDrug	0.5139 ± 0.0015	0.6702 ± 0.0017	0.7672 ± 0.0023	0.0698 ± 0.0009	20.55 ± 0.1632
COGNet	0.5215 ± 0.0019	0.6758 ± 0.0014	0.7712 ± 0.0034	0.0807 ± 0.0023	26.14 ± 0.1061
MoleRec	0.5306 ± 0.0021	0.6843 ± 0.0026	0.7737 ± 0.0023	0.0726 ± 0.0010	20.91 ± 0.1169
DEPOT (**Ours**)	**0.5352 ± 0.0022**	**0.6890 ± 0.0030**	**0.7816 ± 0.0035**	**0.0684 ± 0.0007**	20.62 ± 0.1872

We perform a ten-fold cross-validation to report the mean±variance. ↑ indicates that higher values are desirable, whereas ↓ signifies that lower values are preferable. Bold font is used to highlight the best performance. Avg.# of drugs is commonly reported to convey practicality [12], [27], indicating that the number of drugs cannot be expanded indefinitely to achieve comprehensive coverage. Avg.# of drugs of the ground truth is 19.68.

- *GAMENet [27]:* GAMENet incorporates DDI graph and EHR Graph in the longitudinal modeling process to ensure the effectiveness and safety of recommendations.
- *MICRON [19]:* MICRON leverages residual-based inference to link drug changes and disease changes, enabling incremental learning of new patient characteristics.
- *SafeDrug [6]:* SafeDrug utilizes the atomic architecture of drug molecules for characterization and designs mask matrices to explore the relationship between molecules and diseases. We use the linear adjusting version.
- *COGNet [29]:* COGNet retrieves his or her historical diagnosis and drug recommendations, and mines their relationship with the current diagnosis, which is embedded in Transformer as a plug-in for generative inference.
- *MoleRec [12]:* MoleRec constructs motif embeddings according to the inclusion relationship of molecules and substructures, and designs an attention mechanism to capture their interaction relationship with diseases.

Please Note that these algorithms are categorized as follows, 1) Rule-based: *K-near* 2) Instance-based: *LR, ECC* 3) Longitudinal-based: *RETAIN, LEAP, GAMENet, SafeDrug, MICRON, COGNet, MoleRec*. More detailed distinctions between them and our algorithm are elucidated in Table III. To be fair, we tune the hyper-parameters of each model to achieve the best results.

C. Implementation Details and Evaluation Protocols

Our code has been open-sourced and can be found on GitHub.[2] Regarding the data set, we divided it into three subsets: training set, validation set, and test set as previous [6], [12], [27]. The training set accounts for 2/3 of the data set, while the validation and test sets each comprise 1/6 of the data set. Our model was implemented using PyTorch and trained on an Ubuntu system with a GeForce RTX 3090ti GPU. The initial embeddings were initialized using Kaiming initialization [56] and had a dimension of 64, unless specified otherwise for concatenation operations. To optimize the model during training, we employed the Adam optimizer with a learning rate of 5e-4. We conducted hyperparameter tuning on relevant parameters, including parameter dim within [32,48,64,96], parameter l within [1,2,3,4], parameter T within [2,3,4,5], and parameter τ within [0.05,0.06,0.08,0.1]. See Section IV-E-5 for details.

In evaluating the performance of our recommendation model, we utilize commonly used metrics [27], including Precision-Recall AUC (PR-AUC), Jaccard Similarity Score (Jaccard), and average F1 score (F1). These metrics provide insights into the model's accuracy-related performance. Additionally, we consider the DDI rate as a crucial safety evaluation metric, where

[2]https://github.com/xmed-lab/DrugRec

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TABLE V
PERFORMANCE COMPARISON: MIMIC-IV DATA SET

Method	Jaccard ↑	F1 ↑	PRAUC ↑	DDI ↓	Avg.# of Drugs
K-near	0.3978 ± 0.0023	0.5473 ± 0.0022	0.3917 ± 0.0023	0.0815 ± 0.0013	12.78 ± 0.2203
Logistic Regression	0.4499 ± 0.0033	0.5985 ± 0.0027	0.7259 ± 0.0024	0.0769 ± 0.0008	10.36 ± 0.1111
ECC	0.4350 ± 0.0017	0.5809 ± 0.0020	0.7193 ± 0.0021	0.0681 ± 0.0003	8.76 ± 0.2115
RETAIN	0.4182 ± 0.0020	0.5729 ± 0.0016	0.6775 ± 0.0012	0.0835 ± 0.0004	11.29 ± 0.2007
LEAP	0.4123 ± 0.0022	0.5661 ± 0.0019	0.6026 ± 0.0025	0.0667 ± 0.0005	12.13 ± 0.1530
GAMENet	0.4594 ± 0.0015	0.6126 ± 0.0011	0.7229 ± 0.0020	0.0803 ± 0.0006	18.86 ± 0.3302
MICRON	0.4605 ± 0.0023	0.6132 ± 0.0021	0.6893 ± 0.0022	0.0671 ± 0.0003	13.75 ± 0.2172
SafeDrug	0.4643 ± 0.0035	0.6165 ± 0.0020	0.7051 ± 0.0011	0.0672 ± 0.0004	14.06 ± 0.2615
COGNet	0.4786 ± 0.0010	0.6264 ± 0.0011	0.6963 ± 0.0018	0.0825 ± 0.0002	14.71 ± 0.1105
MoleRec	0.4721 ± 0.0021	0.6242 ± 0.0017	0.7145 ± 0.0015	0.0724 ± 0.0007	14.04 ± 0.2203
DEPOT (**Ours**)	**0.4861 ± 0.0018**	**0.6336 ± 0.0016**	**0.7274 ± 0.0014**	**0.0637 ± 0.0003**	12.77 ± 0.1149

Avg.# of drugs of the ground truth is 12.08.

TABLE VI
PERFORMANCE COMPARISON: EICU DATA SET

Method	Jaccard ↑	F1 ↑	PRAUC ↑	Avg.# of Drugs
K-near	0.1005 ± 0.0005	0.1562 ± 0.0001	0.0838 ± 0.0001	8.46 ± 0.1866
Logistic Regression	0.0779 ± 0.0011	0.3361 ± 0.0024	0.2053 ± 0.0014	9.85 ± 0.7782
ECC	0.0809 ± 0.0015	0.3910 ± 0.0073	0.2041 ± 0.0015	7.45 ± 0.7600
RETAIN	0.1029 ± 0.0042	0.2613 ± 0.0092	0.2319 ± 0.0048	5.16 ± 0.2700
LEAP	0.0925 ± 0.0022	0.4489 ± 0.0081	0.2251 ± 0.0014	9.22 ± 0.2408
GAMENet	0.1061 ± 0.0048	0.4408 ± 0.0083	0.2669 ± 0.0037	10.03 ± 0.3717
MICRON	0.1077 ± 0.0047	0.4471 ± 0.0050	0.2183 ± 0.0050	7.90 ± 0.2894
SafeDrug	0.0913 ± 0.0036	0.4272 ± 0.0064	0.2169 ± 0.0034	10.66 ± 0.2001
COGNet	0.1035 ± 0.0053	0.4549 ± 0.0089	0.2056 ± 0.0041	10.56 ± 0.2341
MoleRec	0.1096 ± 0.0045	0.4473 ± 0.0052	0.2605 ± 0.0027	9.04 ± 0.5995
DEPOT (**Ours**)	**0.1394 ± 0.0047**	**0.4661 ± 0.0096**	**0.2833 ± 0.0016**	12.95 ± 0.2543

Avg.# of drugs of the ground truth is 15.52. Please note that the eicu data set does not include DDI pairs and does not conform to the standards of the current medical coding system, thus can not calculate DDI rate.

TABLE VII
PERFORMANCE COMPARISON: OMOP DATA SET

Method	Jaccard ↑	F1 ↑	PRAUC ↑	Avg.# of Drugs
K-near	0.0443 ± 0.0002	0.6775 ± 0.0048	0.0491 ± 0.0005	1.19 ± 0.1675
Logistic Regression	0.1881 ± 0.0019	0.7199 ± 0.0015	0.3068 ± 0.0046	1.26 ± 0.2288
ECC	0.1846 ± 0.0011	0.6516 ± 0.0039	0.3037 ± 0.0040	1.00 ± 0.2905
RETAIN	0.1894 ± 0.0017	0.4618 ± 0.0017	0.3066 ± 0.0016	0.79 ± 0.3138
LEAP	0.1859 ± 0.0032	0.3555 ± 0.0016	0.3089 ± 0.0045	0.46 ± 0.3059
GAMENet	0.1964 ± 0.0026	0.6067 ± 0.0031	0.3094 ± 0.0041	1.18 ± 0.1818
MICRON	0.1983 ± 0.0011	0.7311 ± 0.0047	0.3143 ± 0.0038	2.52 ± 0.1739
SafeDrug	0.1709 ± 0.0020	0.5697 ± 0.0058	0.2819 ± 0.0033	0.90 ± 0.1175
COGNet	0.1862 ± 0.0017	0.6040 ± 0.0035	0.3008 ± 0.0050	1.11 ± 0.3715
MoleRec	0.1939 ± 0.0028	0.6025 ± 0.0025	0.3104 ± 0.0021	0.89 ± 0.2331
DEPOT (**Ours**)	**0.2130 ± 0.0021**	**0.7493 ± 0.0025**	**0.3148 ± 0.0039**	1.31 ± 0.2387

Avg.# of drugs of the ground truth is 1.58. Please note that the OMOP data set does not include DDI pairs and does not conform to the standards of the current medical coding system, thus can not calculate DDI rate.

a lower DDI rate indicates a safer recommendation in terms of drug-drug interactions. Formally,

$$\text{DDI} = \frac{\sum_{i=1}^{\mathcal{N}_k} \sum_{a,b \in \{j:\hat{\mathbf{o}}_j^{(t)}=1\}} \mathbb{I}\{\mathbf{A}_{a,b}=1\}}{\sum_{i=1}^{\mathcal{N}_k} \sum_{a,b \in \{j:\hat{\mathbf{o}}_j^{(t)}=1\}} 1}, \tag{22}$$

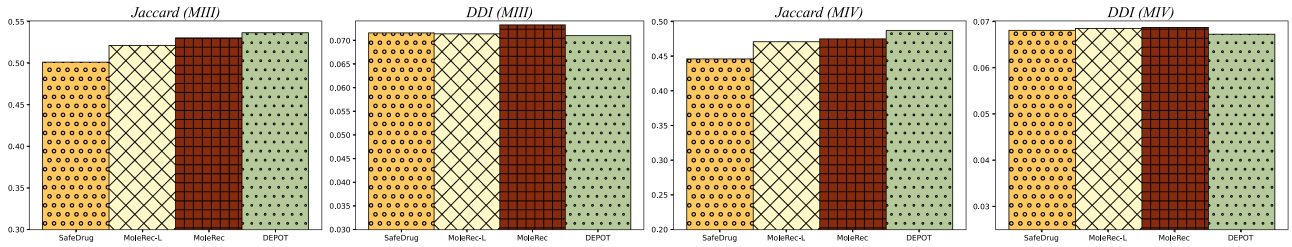
where $\mathbb{I}$ is an indicator function that equals 1 when $\mathbf{A}_{a,b}$ is 1, and 0 otherwise. Furthermore, akin to [6], [12], [27], we track the average number of drugs, a practical metric. While this number may not reflect accuracy or safety, it still holds value for observation [6], [12], [27]. Insufficient drugs may fail to address the user's comprehensive disease requirements, while an excess may escalate financial burdens and prolong the doctor's decision-making time. Additionally, unnecessary additional drugs heighten the risk of drug misuse.

D. Comparison With Baselines

In Tables IV, V, VI, and VII, we present our framework along with the results of all baselines. Overall, *DEPOT* outperforms all other methods across four data sets in terms of accuracy and safety criteria.

Specifically, in terms of accuracy, the K-Near method performs the worst because it relies on a rule-based approach, neglecting the patient's individual factors and the collaborative filtering effects among drugs. Instance-based methods like LR

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(a) Jaccard of MIMIC-III. (b) DDI of MIMIC-III. (c) Jaccard of MIMIC-IV. (d) DDI of MIMIC-IV.

Fig. 6. RECAP decomposition. RECAP is another modern chemical decomposition approach.

TABLE VIII
ABLATION STUDY: MIMIC-III

Method	Jaccard ↑	F1 ↑	PRAUC ↑	DDI ↓	Avg.# of Drugs
DEPOT (**Ours**)	**0.5352 ± 0.0022**	**0.6890 ± 0.0030**	**0.7816 ± 0.0035**	**0.0684 ± 0.0007**	20.62 ± 0.1872
-NS	0.5309 ± 0.0011	0.6841 ± 0.0010	0.7799 ± 0.0012	0.0688 ± 0.0012	21.26 ± 0.2264
-NSM	0.5282 ± 0.0023	0.6808 ± 0.0018	0.7781 ± 0.0015	0.0693 ± 0.0005	20.43 ± 0.1847
-NSG	0.5308 ± 0.0017	0.6844 ± 0.0017	0.7786 ± 0.0008	0.0693 ± 0.0003	21.01 ± 0.1551
-NR	0.5303 ± 0.0015	0.6822 ± 0.0012	0.7798 ± 0.0013	0.0704 ± 0.0006	20.26 ± 0.2134
-NE	0.5298 ± 0.0018	0.6811 ± 0.0014	0.7777 ± 0.0007	0.0696 ± 0.0008	21.07 ± 0.3001
-NH	0.5313 ± 0.0013	0.6856 ± 0.0009	0.7797 ± 0.0014	0.0696 ± 0.0009	20.45 ± 0.3339
-NST	0.5299 ± 0.0038	0.6841 ± 0.0016	0.7778 ± 0.0034	0.0690 ± 0.0008	20.46 ± 0.2995
-NV	0.5296 ± 0.0018	0.6839 ± 0.0014	0.7778 ± 0.0019	0.0690 ± 0.0009	20.74 ± 0.3100

(1) DEPOT-NS is not aware of motifs collaborations and uses atomic embeddings for graph pooling. (2) DEPOT-NSM utilizes transformer to encode smiles strings, treating them as drug representations. (3) DEPOT-NSG utilizes the molecule graph and employs GCN layers to extract drug representation. (4) DEPOT-NR removes the repeat module. (5) DEPOT-NE removes the explore module. (6) DEPOT-NH has no historical DDI considerations. (7) DEPOT-NST does not add structural information in the graph transformer. (8)DEPOT-NV employs a linear layer to substitute the distribution encoder in eq. 9.

and ECC show greater improvements in recall and accuracy compared to K-Near, with a nearly 4.5% improvement in jaccard indicators. However, due to the lack of consideration for the patient's historical information, they cannot fully capture the patient's state, resulting in significantly lower performance compared to *DEPOT*. Regarding longitudinal-based baselines, RETAIN and LEAP exhibit weaker performance in comparison with SafeDrug and COGNet on the MIMIC data sets as they ignore topological information in molecular characterization and high-order similarity. MoleRec, which also models from motifs, is the strongest baseline. Nonetheless, its effectiveness is still inferior to ours in jaccard, with 1%, 3%, 27% and 9% lower performance observed across four data sets. Similar decline has been observed across other metrics as well. This disparity arises from its failure to account for the structural connections between motifs and the dynamic relationship between motifs and disease progression.

In terms of safety, the baselines that include DDI regularization, such as GAMENet, MICRON, and COGNet, outperform the baselines that do not consider security criteria, such as K-near, LR, and ECC, on both MIMIC data sets. Among them, SafeDrug achieves the most competitive DDI rate, which stems from its strict principle of sacrificing recommendation accuracy, that is, only optimizes DDI when the target DDI is exceeded. In contrast, *DEPOT* optimizes the DDI loss in a non-linear form and utilizes historical DDI rates to stabilize training, enabling exploration of a broader optimization space. While MoleRec's accuracy metrics on MIMIC-III are comparable to ours, its DDI rate is 6% higher than ours, potentially resulting in more adverse drug effects. This comprehensive approach allows *DEPOT* to achieve the superior performance in terms of safety, further distinguishing it from other baselines.

In terms of practicality, it is noteworthy that our algorithm ensures only 1 or 2 deviations from ground truth in the average number of drugs. In contrast, methods such as GAMENet and COGNet tend to enhance accuracy by incorporating a larger number of drugs in MIMIC data sets, which carries the potential risk of increased side effects and longer decision times. Moreover, within the demanding contexts of eICU and OMOP, marked by truncated sequences and a broader spectrum of drug combinations, conventional algorithms such as MoleRec and RETAIN often advocate for fewer drugs, predominantly leaning on previously seen options. Conversely, our algorithm advocates for a broader spectrum of drugs, guided by our more comprehensive dual perspective of repetition and exploration.

In terms of data sets, the performance of all algorithms on the MIMIC-IV data set shows a slight decrease compared to MIMIC-III, attributed to the data set's greater diversity and heterogeneity in diagnosis and treatment options, as well as its more complex and challenging drug-disease patterns. The eICU and OMOP data sets present greater challenges than MIMIC, as evidenced by more complex drug combinations, shorter longitudinal data spans, and reduced supervisory signals. These complexities pose significant difficulties for many algorithms, especially for Logistic Regression and SafeDrug, their performance is surpassed by K-near in jaccard. COGNet and MoleRec exhibit decreased rank in these two data sets compared to previous results. COGNet's limitations stem from inadequately sized EHR and absence of DDI graphs, impeding its ability to handle complex drug interactions effectively. On

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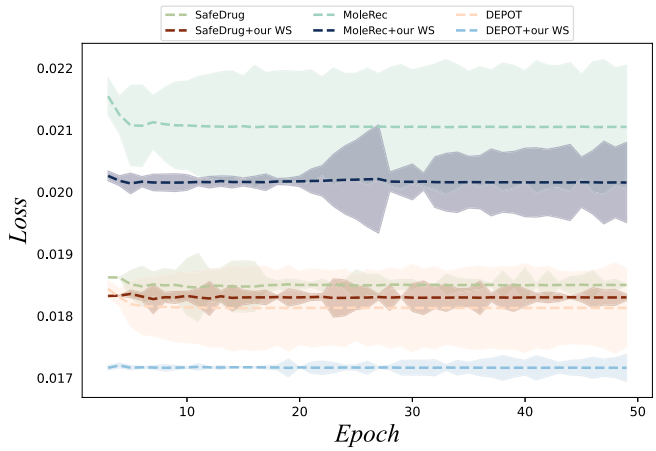


Fig. 7. Plug-in DDI Loss. + our WS means using our weighted strategy. The shaded area is the range of the error bar.

the other hand, MoleRec is biased towards previously encountered drugs when confronted with extensive drug combinations due to its deficiency in essential exploration mechanisms. In contrast, our algorithm consistently achieves superior performance, outperforming the best baseline across all metrics.

To sum up, *DEPOT* demonstrates superior performance in terms of accuracy, safety, and practicality, owing to its fine-grained utilization of motifs and enhanced nonlinear weighting strategy.

E. Discussion

To acquire comprehensive insights into the functions and effectiveness of our design, we conduct diverse discussions.

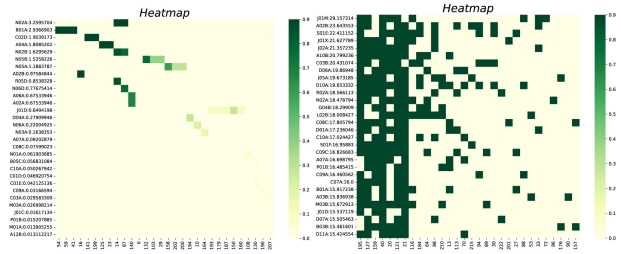
1) Diverse Motif-Tree Construction: To further assess the algorithm's stability, we applied the RECAP decomposition [57] to reconstruct the motif-tree and conducted comparisons with several competitive algorithms. RECAP mimics the forward synthesis process in the laboratory, executing reverse operations to transform molecules into a set of fragment structures common to biologically active substances, which is widely applied in molecular decomposition [57]. The results, presented in Fig. 6, demonstrate that our algorithm consistently achieves favorable experimental outcomes even under the RECAP decomposition method. It indicates that our approach can effectively handle variations in data decomposition techniques, ensuring consistent performance. This versatility enhances the algorithm's practical utility and suggests its potential for application in different real-world scenarios. Moreover, considering the possibility of future advancements in semantic decomposition methods, our algorithm holds promise for further improvement. By leveraging more sophisticated semantic decomposition techniques, our algorithm has the potential for pluggable enhancements, enabling it to adapt to evolving decomposition methodologies and potentially achieve even better performance.

2) DDI Loss: Fig. 7 illustrates the impact of integrating the historical DDI weight determination strategy into SafeDrug (linear), MoleRec (nonlinear), and *DEPOT* (nonlinear) on their DDI loss. We additionally incorporate error bars to illustrate

TABLE IX
EXAMPLE RECOMMENDATION RESULTS

Method	Recommended Drug Set
Ground-Truth	**TP:** ['A06A', 'N02B', 'B01A', 'A01A', 'C02D',
Num: 10	'A04A', 'N02A', 'A12B', 'M03B', 'C08C']
SafeDrug	**TP:** ['N02A', 'A12B', 'A04A', 'A01A', 'B01A',
Num:12	'N02B', 'A06A']
Recall:0.700	**FN:** ['C02D', 'C08C', 'M03B']
Precision:0.583	**FP:** ['B05C', 'N05B', 'A12C', 'A02B', 'D04A']
MoleRec	**TP:** ['N02A', 'C02D', 'N02B', 'B01A', 'A06A',
Num:10	'A12B', 'A01A']
Recall:0.700	**FN:** ['C08C', 'M03B', 'A04A']
Precision:0.700	**FP:** ['A12C', 'A02B', 'B05C']
DEPOT	**TP:** ['B01A', 'A04A', 'A12B', 'A01A', 'C02D',
Num:11	'A06A', 'N02B', 'N02A']
Recall:0.800	**FN:** ['M03B', 'C08C']
Precision:0.727	**FP:** ['A02B', 'B05C', 'A12C']

True positives (TP), false negatives (FN), and false positives (FP).



(a) Drug-centric. (b) Patient-centric.

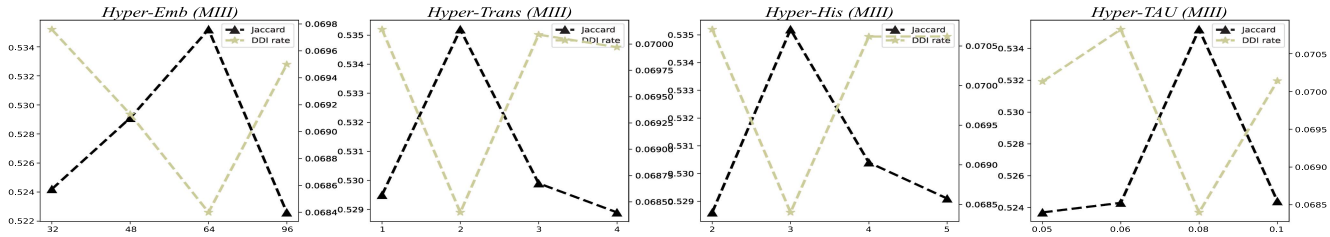
Fig. 8. Heat map. The color bar denotes the Drug-centric/Patient-centric motif selection probability, with darker colors indicating higher probabilities. We showcase the top 30 motifs and drugs, excluding very low-scoring ones to enhance visualization clarity. The x-axis of the heatmap corresponds to the motif index, while the y-axis represents the drug code along with its score. This score is derived by multiplying the probabilities (Drug-centric/Patient-centric) with the motifs present in the drug.

fluctuations in curve stability. On the one hand, the evidence in the figure demonstrates that the three algorithms exhibit varying degrees of reduction in the lower bound of DDI loss convergence, thus confirming that the incorporation of historical information ensures safer recommendations. On the other hand, the integration of historical DDI enables the three models to achieve a more stable training process, resulting in a smoother loss curve and smaller error bar range. This stability contributes to more reliable and consistent predictions, ultimately improving the accuracy and robustness of *DEPOT*.

3) Ablation Studies: We further compare *DEPOT* with several ablation variants to demonstrate the effectiveness of different sub-modules. For fairness, we maintain consistent settings across all experiments except for the specific ablation module.

In Table VIII, we present the results of different variant methods, showcasing their performance across various metrics. Among the ablation variants, the *DEPOT* stands out as the top performer across all metrics. On the other hand, the *DEPOT*-NE method demonstrates the notable decline. This is attributed to its limited exploration of the relationships between previously interacted drug motifs, leading to a narrowed recommendation range. *DEPOT*-NR is less degraded compared to *DEPOT*-NE, as it benefits from the collaborative filtering of diseases and drugs

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(a) The impact of dim. (b) The impact of l. (c) The impact of T. (d) The impact of τ.

Fig. 9. Hyper-parameter tests.

through a drug-centric module. *DEPOT*-NS, *DEPOT*-NSM, and *DEPOT*-NSG represent three drug modeling methods employed to replace motif-level aggregation in Section III-D. Among them, *DEPOT*-NSM exhibited the most pronounced decline, primarily due to SMILES' inability to capture subtle topology differences between drugs, operating primarily at the inclusion level. While *DEPOT*-NS and *DEPOT*-NSG can capture drug correlations at a finer granularity of atoms, they lack direct evidence of drug functionality, resulting in certain performance reduction. Comparing *DEPOT*-NH to *DEPOT*, both models exhibit similar accuracy values. However, Drug-NH slightly lags behind *DEPOT* in terms of DDI. This discrepancy can be attributed to the impact of DDI outliers in the model. The performance degradation observed in *DEPOT*-NST results from the neglect of topological structure information during the aggregation process. This oversight may lead to a misunderstanding of motif combinations within the model. *DEPOT*-NV fails to mimic the randomness and uncertainty of disease progression in medical scenarios, leading to suboptimal and unstable results. Overall, each module in our design plays a crucial role in the effectiveness of the *DEPOT* model. The superiority of the *DEPOT* and the relatively weaker performance of the ablation variants highlight the significance of the individual modules and the comprehensive design of our framework.

4) Case Study: To showcase the superiority of *DEPOT* and provide an intuitive understanding, we present the recommendation of *DEPOT*, SafeDrug, and MoleRec for an anonymous patient. Analyzing the results in Table IX, it becomes evident that *DEPOT* surpasses other methods in both recall and precision. This exceptional performance can be attributed to the fact that *DEPOT* considers the intricate functional collaboration between drug motifs and their relevance to disease processes during the recommendation process. To minimize the occurrence of potent side effects, certain drugs, specifically *M03B* and *C08C*, have been omitted from the recommended results of *DEPOT*. In contrast, SafeDrug overlooks potential drugs due to excessively strict safety guidelines, while MoleRec fails to consider possible drug reuse due to its lack of a dynamic perspective. *We recognize that doctors' levels of experience and expertise can vary, resulting in different prescribed drugs for patients. As a consequence, there is no absolute gold standard for comparison. However, the findings presented strongly suggest that DEPOT holds promise as an advanced drug recommender system.*

We also present heatmaps illustrating the scores of drug motifs based on both drug-centric and patient-centric strategies. As shown in Fig. 8, we observe a robust correlation between motif preference and drug selection, evidenced by the inclusion of motifs with high attention scores in the majority of selected drugs. On the other hand, model recommendation depends on the synergistic action of both modules, as no single module comprehensively covers all recommended outcomes.This demonstrates the effectiveness of our design.

5) Hyper-Parameter Tests: To obtain the optimal model performance and observe the impact of key parameters on it, in this subsection, we conduct hyper-parameter tests on embedding size, the number of transformer layers, historical length, and temperature.

Embedding Size dim. To examine the impact of embedding dimension on the performance of our proposed model, we conduct scaling experiments by varying the embedding size. The results are presented in Fig. 9(a).

The findings demonstrate that *DEPOT* initially experiences improvement as the embedding dimension increases. This improvement continues until reaching a peak when the dimension is set to 64. However, when the dimension exceeds 64, there is a significant drop in performance. This decline can be attributed to the introduction of noise and the potential for overfitting in the model when the dimensionality becomes too large. Experimentally, we set $dim = 64$.

Number of Transformer Layers l. The number of Transformer layers in our model plays a crucial role in capturing higher-order collaboration between motifs. Fig. 9(b) illustrates the results of our experiments, indicating that the best performance is achieved when $l = 2$ (the optimal number of layers). When the number of layers is too small, the model may struggle to effectively capture the structural relationships between motifs. This limitation hampers the model's ability to understand complex interactions and may result in suboptimal performance. Conversely, when the number of layers is excessively large, the performance improvement may reach saturation, making further increases in the number of layers unnecessary and inefficient.

Historical Length T. When the historical length is longer, the calculated historical DDI rate is more stable and accurate. The calculation of the average value is based on a larger number of data points, providing a better reflection of the trends in historical DDI interactions. However, there may be some potential issues

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with excessively long historical lengths, such as increased computational complexity and data storage requirements. Moreover, longer historical lengths may include outdated or less representative data, which can have a negative impact on the accuracy of the results. As shown in Fig. 9(c), we observe better performance when $T = 3$.

Temperature τ. τ controls the nonlinearity of the tanh function, which affects the level of constraint on safety objective in (21). A smaller value of τ causes the curve to rise more rapidly when it surpasses the target DDI, thereby assigning higher weight to DDI optimization. Conversely, a larger value of τ results in a more gradual curve and less emphasis on DDI optimization. As illustrated in Fig. 9(d), our model achieves superior performance on both accuracy and safety objectives when $\tau = 0.08$.

V. Conclusion

In this study, we present *DEPOT*, a drug recommendation framework that harnesses drug motifs to deliver precise and secure prescription recommendations. Our framework incorporates molecular decomposition and structure-aware graph transformers to sense topology knowledge and motif collaborations, capturing drug functionality. Additionally, we introduce two modules, repeat encoder and explore encoder, to capture the dynamic relationship between drug motifs and disease progression. Moreover, we integrate historical DDI information in a non-linear fashion to ensure recommendation stability and safety criteria. Through comprehensive comparison and robustness testing on four large medical data sets, we demonstrate that *DEPOT* showcases improved accuracy, safety and practicality, thereby assisting healthcare professionals in making informed decisions and paving the way for further advancements in the field of drug recommendation.

Nonetheless, despite these remarkable results, there are still avenues for further exploration. First, it is crucial to address patient privacy concerns and implement appropriate privacy protection measures during the recommendation process. Second, we aim to extend our framework to other healthcare recommendation, such as diagnosis and treatment recommendations, to cater to a broader range of medical needs.

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